Exact K2P and K3P Tree–Theta-Trinet Collisions

Technical summary

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Exact finding. A binary semi-directed strict level-two theta trinet with a genuine nontrivial 3-blob shares an exact K2P distribution with a three-star tree. This is a counterexample to Lemma 5.6 and the K2P part of Corollary 5.8 in arXiv:2607.12919v2. Version 3 removes those claims, identifies the leaf-order obstruction, and leaves high-level K2P and K3P distinguishability open. Because K2P $\subset$ K3P, the K2P collision alone answers both questions negatively. A separate quartic construction gives an exact theta realization whose network parameter is outside every globally character-relabelled K2P specialization. Its shared distribution is openly identified as globally character-relabelled K2P; full K3P rank supplies nearby shared distributions outside all globally character-relabelled K2P models. Here "globally" means that the same permutation of the three nonidentity characters is used on every edge and leaf; edge-dependent relabellings are excluded. The corrected paper's JC and level-one results are unaffected.

Common topology. Use

$$\rho \to 1, \rho \to u,\ u \to p, q,\ p, q \to r_2,\ p, q \to r_3,\ r_2 \to 2,\ r_3 \to 3,$$

with both inheritance probabilities 1/2. Root suppression gives paths $p - u - q$, $p - r_2 - q$, $p - r_3 - q$ and three incident leaf edges. For the explicit symmetric witnesses, U labels $u \to p$, V labels $u \to q$, S labels both $p \to r_2, p \to r_3$, and T labels both $q \to r_2, q \to r_3$; the other four arcs carry K. The rank and implicit-function calculations instead use independent parameters on all nine effective semi-directed edges and both inheritance probabilities.

Simple exact K2P witness. In order (A, C, G, T), assign

$$K = (1, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), \quad U = (1, \tfrac{4}{5}, \tfrac{19}{30}, \tfrac{4}{5}), \quad V = (1, \tfrac{7}{240}, \tfrac{239}{360}, \tfrac{7}{240}),$$
$$S = (1, \tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}), \quad T = (1, \tfrac{1}{3}, \tfrac{1}{27}, \tfrac{1}{3}).$$

Use K on the four common K-edges $\rho \to 1, \rho \to u, r_2 \to 2, r_3 \to 3$, with the explicit U, V, S, T placement above. Every transition entry is positive; the smallest is 1/120. For $x = y + z$, these common edges contribute $K_x K_{y+z} K_y K_z = K_x^2 K_y K_z$. The four retained-parent choices $(p, p), (p, q), (q, p), (q, q)$ give, respectively, the four terms in

$$M_{y,z} = \tfrac{1}{4}(S_y S_z U_x + S_y T_z U_y V_z + T_y S_z U_z V_y + T_y T_z V_x).$$

With $\eta = \sqrt{71}$,

$$P = (1, \tfrac{151}{36\eta}, \tfrac{107}{162}, \tfrac{151}{36\eta}), \quad R = (1, \tfrac{\eta}{40}, \tfrac{31}{120}, \tfrac{\eta}{40}),$$

and exact arithmetic gives $M_{y,z} = P_{y+z} R_y R_z$. The tree vectors are

$$\alpha = (1, \tfrac{151\eta}{10224}, \tfrac{107}{648}, \tfrac{151\eta}{10224}), \quad \beta = \gamma = (1, \tfrac{\eta}{80}, \tfrac{31}{240}, \tfrac{\eta}{80}).$$

All 64 Fourier and pattern probabilities agree exactly; the minimum is $1188799/79626240 = 0.01492973924\ldots$. The source invariant Q is zero. A graph-based verifier literally deletes the two unselected reticulation arcs, reconstructs the four monomials, and independently confirms all 64 probabilities by ordinary-state Markov pruning.

Proof diagnosis. The level-one sign factorization is valid after placing the active reticulation leaf in a designated position. In the level-two induction a child can retain a reticulation adjacent to another leaf. Positivity after a child-specific relabelling does not imply positivity of the fixed-order child invariant in the parent expansion. Version 3 now records this obstruction. On the exact edgewise continuous-time witness, certified bounds give $-1.9200 \times 10^{-9} < Q < -1.9199 \times 10^{-9}$ in the parent order and $3.4284 \times 10^{-9} < Q < 3.4286 \times 10^{-9}$ after the favorable swap. A separate fully specified rational theta point has $Q < 0$ in all six orders.

K2P geometry and edgewise continuous time. Here "edgewise" means that every edge separately has a positive-rate symmetric generator; no common generator, rate ratio, molecular clock, or global temporal compatibility is imposed. The affine K2P space has dimension 9 and the tree model dimension 6. At the simple collision, a selected 9×9 Jacobian minor of the fixed theta map has determinant

$$-\frac{7^2 11^2 19\, 107\, 151^2\, 15013}{2^{60} 3^{25} 5^{10}} \neq 0.$$

It also has rank 9 at the edgewise strictly continuous-time collision. Hence the local semi-directed theta collision locus has dimension 17 and codimension 3, and it fibers submersively over nearby tree distributions with 11-dimensional fixed-output fibers. Generic theta parameters are not tree-equivalent, but full pointwise tree–theta distinguishability fails. An exact symmetric ansatz contains a smooth local six-dimensional semialgebraic family of positive collisions. The edgewise continuous-time witness satisfies $g > s^2$ on every edge, has minimum eigenvalue $1/16$ and minimum rate margin $11/900$, and is verified by direct Markov pruning.

K3P symmetry breaking and local geometry. Let $h = 5^{-1/4}$. Use the same fixed edge placement and the same four-switching matrix M, now with

$$U = (1, h/3, h, 1/3),\ V = (1, h, h/3, 1/3),\ S = (1, 3h^2/4, 1/4, 3/10),\ T = (1, 1/4, 3h^2/4, 3/10).$$

This mixture factors with

$$B = (1, h^2/2, h^2/2, h^2/2), \quad P = (1, (5h^3 + h)/4, (5h^3 + h)/4, h^2).$$

The U edge has three pairwise-distinct nontrivial eigenvalues, so this network parameter is outside every globally character-relabelled K2P specialization. With $a = (5h^3 + h)/16$ and $t = h^2/4$, the comparison tree has $\alpha = (1, a, a, t)$ and $\beta = \gamma = (1, t, t, t)$, so its shared distribution is in the $C = G$ globally character-relabelled K2P model and is not JC. The fixed theta map has rank 15 with

$$\det J_* = \frac{h(10h^2 + 1)}{2^{61} 3^4 5^{14}} > 0.$$

Its local collision locus has dimension 23 and codimension 6 and fibers over the nine-dimensional tree model with 14-dimensional fixed-output fibers. Within the locally covered K3P tree neighborhood, the complement of the three six-dimensional globally character-relabelled K2P strata is relatively open and dense, and every distribution in that complement is an observably genuine K3P collision. A real-analytic implicit-function argument moves the construction into the edgewise cone where all three K3P rate classes are positive, and the same local conclusion holds there.

Algebraic and all-leaf consequences. The full-rank minors make the complexified fixed-theta images Zariski dense in the effective affine spaces $\mathbb{A}^9$ and $\mathbb{A}^{15}$. Thus no additional polynomial equality vanishes on an entire fixed-theta image; stochastic and continuous-time inequalities remain, and the tree varieties still have invariants. If any internal vertex of any labelled unrooted binary tree on $n \geq 3$ leaves is replaced by one theta blob, attaching the same three conditional Markov subtrees preserves equality. This gives strict-interior and edgewise continuous-time collisions on every such topology. Equivariant injective JC subtree kernels also preserve the absence of every globally character-relabelled K2P symmetry for the nearby genuine K3P collisions. Each comparison-tree attachment can be subdivided into two strict half-time edges; the theta rooting uses only the compatible terminal-1 side. This is a one-blob theorem, not a multi-blob composability result. The theta admits no tree-child rooting, so none of these statements contradicts results restricted to tree-child or strongly tree-child level-two networks.

Status and replay. The canonical manuscript, exact certificates, and verifier package are available in the versioned GitHub snapshot (v1.2.2). With Python 3.10 or newer, run `python3 verify_k2p_displayed_trees.py` for the graph reconstruction and direct-pruning audit, `python3 src/verify_k2p_four_leaf_graft.py` for the exact $n = 4$ regression, or `python3 verify.py` for every check; the latter ends with `ALL EXACT CHECKS PASSED`.